

A Course on Rough Paths

Solutions to rough differential equations

Woojoo Shim

May 15, 2020

Department of Mathematical Sciences
HYKE weekly seminar

1. Review
2. Solutions to rough differential equations
3. Regularity of the Itô-Lyons map
4. Summary

Review

- How to solve controlled ODEs of the form

$$\dot{Y}_t = f_0(Y_t) + f(Y_t)\dot{X}_t$$

when the input $\dot{X}_t$ is assumed to be a Brownian white noise?

- A special non-smooth theory is needed to solve such ODEs in the following sense:

Proposition 1.1

There exists no separable Banach space $\mathcal{B} \subset \mathcal{C}([0, 1])$ with the following properties:

1. Sample paths of Brownian motions lie in $\mathcal{B}$ almost surely
2. The map $(f, g) \mapsto \int_0^\cdot f(t)\dot{g}(t)dt$ defined on smooth functions extends to a continuous map from $\mathcal{B} \times \mathcal{B}$ into $\mathcal{C}([0, 1])$.

⇒ There is no universal, reasonable way to extend the ODE theory for such systems.

- We may extend the ODE theory to several different ways! (Itô, Stratonovich, etc ...)
- The possible extensions are determined by the 'designation' of

$$B \mapsto \int_0^\cdot B^i(t) \dot{B}^j(t) dt$$

for Brownian motion B .

Introduction

- We are mainly interested to the **Itô map** denoted by S , which maps a Brownian motion $B(\omega)$ to corresponding solution $Y(\omega)$.
- The Itô map can be decomposed to the '**designation**' map Ψ and the **Itô-Lyons map** $\hat{S}$. Namely,

$$\Psi : B(\omega) \mapsto (B, \mathbb{B})(\omega), \quad \hat{S} : (B, \mathbb{B})(\omega) \mapsto Y(\omega),$$

where

$$\mathbb{B}^{ij}(s, t) =: \int_s^t (B^i(r) - B^i(s)) dB^j(r).$$

- The Itô-Lyons map has some kind of continuity.

What is rough path?

- A reasonable extension of the designation map Ψ to a large class of functions.
- For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, define the space of α -Hölder rough paths over Banach space V , in symbols $\mathcal{C}^\alpha([0, T], V)$, as those pairs $(X, \mathbb{X}) =: \mathbf{X}$ such that

$$\|X\|_\alpha := \sup_{s \neq t \in [0, T]} \frac{|X_{s,t}|}{|t-s|^\alpha} < \infty, \quad \|\mathbb{X}\|_{2\alpha} := \sup_{s \neq t \in [0, T]} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} < \infty,$$

and satisfy the **Chen's relation**:

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}.$$

- Itô and Stratonovich Brownian motions can be regarded as rough paths.

Integration against rough path

- Recall that the Young integral is defined as

$$\int_s^t Y_r dX_r = \lim_{|\mathcal{P}| \rightarrow 0} \sum_{[u,v] \in \mathcal{P}} Y_u X_{u,v},$$

where $\mathcal{P}$ denotes a partition of $[s, t]$.

- The above sum converges if $X \in \mathcal{C}^\alpha$, $Y \in \mathcal{C}^\beta$ and $\alpha + \beta > 1$.
- However, if we postulate the values $\mathbb{X}_{s,t}$ and if Y “looks like X ” (e.g. $Y = F(X)$ for regular F), we may relax the above Hölder regularity condition to define the integral.

Integration against rough path

Definition (Controlled rough path)

Given a path $X \in \mathcal{C}^\alpha([0, T], V)$, we say that $Y \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, W))$ is **controlled by X** if there exists $Y' \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, \mathcal{L}(V, W)))$ so that the remainder term R^Y given through the relation

$$Y_{s,t} = Y'_s X_{s,t} + R^Y_{s,t},$$

satisfies $\|R^Y\|_{2\alpha} < \infty$. This defines the space of controlled rough paths,

$$(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], \mathcal{L}(V, W)).$$

- Y' is called the **Gubinelli derivative** of Y with respect to X .
- $\|Y, Y'\|_{X, 2\alpha} := \|Y'\|_\alpha + \|R^Y\|_{2\alpha}$ is a semi-norm of $\mathcal{D}_X^{2\alpha}$.
- $\mathcal{D}_X^{2\alpha}$ is a Banach space under the norm

$$(Y, Y') \mapsto |Y_0| + |Y'_0| + \|Y, Y'\|_{X, 2\alpha}.$$

Integration against rough path

Definition (Integration against rough path)

Let $T > 0$ and $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T], V)$ for some $\alpha > \frac{1}{3}$, and let $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], \mathcal{L}(V, W))$. Then, we define

$$\int_s^t Y d\mathbf{X} := \lim_{|\mathcal{P}| \rightarrow 0} \sum_{[u,v] \in \mathcal{P}} (Y_u X_{u,v} + Y'_u \mathbb{X}_{u,v}).$$

Integration against rough path

Theorem (Gubinelli)

Let $T > 0$ and $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T], V)$ for some $\alpha > \frac{1}{3}$, and let $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], \mathcal{L}(V, W))$. Then, there exists a constant C depending only on α such that

- For every pairs s, t , one has the bound

$$\left| \int_s^t Y_r d\mathbf{X}_r - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right| \leq C_\alpha \left(\|X\|_\alpha \|R^Y\|_{2\alpha} + \|X\|_{2\alpha} \|Y'\|_\alpha \right) |t - s|^{3\alpha}.$$

- The map from $\mathcal{D}_X^{2\alpha}([0, T], \mathcal{L}(V, W))$ to $\mathcal{D}_X^{2\alpha}([0, T], W)$ given by

$$(Y, Y') \mapsto (Z, Z') := \left(\int_0^\cdot Y_t d\mathbf{X}_t, Y \right)$$

is a continuous linear map between Banach space with bound

$$\|Z, Z'\|_{X, 2\alpha} \leq \|Y\|_\alpha + \|Y'\|_\infty \|\mathbb{X}\|_{2\alpha} + CT^\alpha \left(\|X\|_\alpha \|R^Y\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Y'\|_\alpha \right).$$

Solutions to rough differential equations

We want to solve **rough** differential equation(RDE)

$$Y_t = \xi + \int_0^t f(Y_s) d\mathbf{X}, \quad (1)$$

for sufficiently regular f and **rough** path $\mathbf{X} = (X, \mathbb{X})$.

Idea:

- We consider a mapping $\mathcal{M}_T : \mathcal{D}_X^{2\alpha} \rightarrow \mathcal{D}_X^{2\alpha}$ given by

$$\mathcal{M}_T(Y, Y') := \left(\xi + \int_0^\cdot \Xi_s d\mathbf{X}_s, \Xi \right),$$

and find its **fixed point**.

- If necessary, we assume $T \in (0, 1]$ sufficiently small to apply the Contraction mapping theorem, and iterate the procedure as much as we can.

Composition with regular functions

Lemma 1 (Estimate on a controlled rough path)

Let W and $\bar{W}$ be two Banach space and let $\varphi \in \mathcal{C}_b^{2\alpha}(W, \bar{W})$. Let furthermore $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], W)$ for some $X \in \mathcal{C}^\alpha$ with

$$|Y'_0| + \|Y, Y'\|_{X, 2\alpha} \leq M, \quad M \geq 1.$$

Then, $(\phi(Y)_t, \phi(Y)'_t) := (\phi(Y_t), D\phi(Y_t)Y'_t)$ is in $\mathcal{D}_X^{2\alpha}([0, T], \bar{W})$ and there exists a constant C depending only on $T > 0$ and $\alpha > \frac{1}{3}$ such that

$$\|\varphi(Y), \varphi(Y)'\| \leq C_{\alpha, T} M \|\varphi\|_{\mathcal{C}_b^{2\alpha}} (1 + \|X\|_\alpha)^2 (|Y'_0| + \|Y, Y'\|_{X, 2\alpha}).$$

At last, $C_{\alpha, T}$ can be chosen uniformly over $T \in (0, 1]$.

Contraction mapping theorem

Contraction mapping theorem (Banach fixed point theorem)

Let (X, d) be a non-empty complete metric space, and suppose that there exists a map $T : X \rightarrow X$ and $r \in [0, 1)$ such that

$$d(Tx, Ty) \leq rd(x, y), \quad \forall x, y \in X.$$

Then, T admits a unique fixed point x_0 in X .

Existence and uniqueness

Theorem (Existence and uniqueness)

For given $\xi \in W$, $f \in \mathcal{C}^3(W, \mathcal{L}(V, W))$ and $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\beta(\mathbb{R}_+, V)$ with $\beta \in (\frac{1}{3}, \frac{1}{2})$, there exists a unique element $(Y, Y') \in \mathcal{D}_X^{2\beta}([0, 1], W)$ such that $Y' = f(Y)$ and

$$Y_t = \xi + \int_0^t f(Y_s) d\mathbf{X}_s, \quad t < \tau,$$

for some $\tau > 0$. Moreover, if $f \in \mathcal{C}_b^3$, the solution Y exists global in time.

Proof: (Step 1) (Basic estimates on $\mathcal{M}_T$) First, for any α with $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\beta \subset \mathcal{C}^\alpha$, $\frac{1}{3} < \alpha < \beta < \frac{1}{2}$ and $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, we can apply the Lemma 1 to get

$$(\Xi, \Xi') := (f(Y), f(Y)') = (f(Y), Df(Y)Y') \in \mathcal{D}_X^{2\alpha}([0, T], \mathcal{L}(V, W)).$$

Existence and uniqueness

Then, as we assume $T \leq 1$ and restricting from $[0, 1]$ to $[0, T]$, we apply the Gubinelli's theorem to (Ξ, Ξ') to define the map

$$\mathcal{M}_T : \mathcal{D}_X^{2\alpha}([0, T], W) \rightarrow \mathcal{D}_X^{2\alpha}([0, T], W),$$
$$\mathcal{M}_T(Y, Y') := \left(\xi + \int_0^\cdot \Xi_s d\mathbf{X}_s, \Xi \right),$$

so that any fixed point of $\mathcal{M}_T$ is a possible candidate of the RDE solution in $\mathcal{D}_X^{2\alpha}([0, T], W)$. But since $\mathbf{X} \in \mathcal{C}^\beta$ and $\beta < \frac{1}{2} < \frac{2}{3} < 2\alpha$, we have

$$|Y_{s,t}| \leq \|Y'\|_\infty |X_{s,t}| + \|R^Y\|_{2\alpha} |t - s|^{2\alpha} \implies Y \in \mathcal{C}^\beta.$$

Moreover, $Y' = \Xi = f(Y) \in \mathcal{C}^\beta$ for the fixed point (Y, Y') and also $R^Y \in \mathcal{C}_2^{2\beta}$, since $\mathbb{X} \in \mathcal{C}_2^{2\beta}$, $2\beta < 1 < 3\alpha$ and

$$\begin{aligned} |R_{s,t}^Y| &= |Y_{s,t} - Y'_s X_{s,t}| = \left| \int_s^t (f(Y_r) - f(Y_s)) d\mathbf{X}_r \right| \\ &\leq \|\Xi'\|_\infty |\mathbb{X}_{s,t}| + O(|t - s|^{3\alpha}). \end{aligned}$$

(See (8.3) in the textbook and use Gubinelli's theorem)

Existence and uniqueness

Therefore, for every $\frac{1}{3} < \alpha < \beta < \frac{1}{2}$, a fixed point of

$$\mathcal{M}_T : \mathcal{D}_X^{2\alpha}([0, T], W) \rightarrow \mathcal{D}_X^{2\alpha}([0, T], W),$$

is **indeed contained in** $\mathcal{D}_X^{2\beta}([0, T], W)$. On the other hand, for every (Y, Y') with $(Y_0, Y'_0) = (\xi, f(\xi))$, the same is true for $\mathcal{M}_T(Y, Y') = (\xi + \int_0^\cdot \Xi_s d\mathbf{X}_s, \Xi)$. Since $\mathcal{D}_X^{2\alpha}([0, T], W)$ is a Banach space under the norm $(Y, Y') \rightarrow |Y_0| + |Y'_0| + \|Y, Y'\|_{X, 2\alpha}$, any closed unit ball in the affine subspace

$$\{(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], W) : Y_0 = \xi, Y'_0 = f(\xi)\}$$

is a complete metric space under the induced metric. We denote $\mathcal{B}_T$ as such unit ball centred at $t \mapsto (\xi + f(\xi)X_{0,t}, f(\xi))$.

Existence and uniqueness

Note that the distance between $(Y, Y') \in \mathcal{B}_T$ and the center $t \mapsto (\xi + f(\xi)X_{0,t}, f(\xi))$ is:

$$\begin{aligned} & |Y_0 - \xi| + |Y'_0 - f(\xi)| + \|Y - (\xi + f(\xi)X_{0,\cdot}), Y' - f(\xi)\|_{X,2\alpha} \\ &= \|Y - f(\xi)X_{0,\cdot}, Y' - f(\xi)\|_{X,2\alpha} \\ &= \|Y' - f(\xi)\|_\alpha + \|R^{Y-f(\xi)X_{0,\cdot}}\|_{2\alpha} \\ &= \|Y'\|_\alpha + \|R^Y\|_{2\alpha} = \|Y, Y'\|_{X,2\alpha}. \end{aligned}$$

Now, for all $(Y, Y') \in \mathcal{B}_T$, one has the bound

$$|Y'_0| + \|Y, Y'\|_{X,2\alpha} \leq \|f(\xi)\| + 1 =: M \in [1, \infty).$$

Unless f is assumed to be $\mathcal{C}_b^3$, we write

$$\|f\|_\infty := \|f\|_{\mathcal{C}_b(\pi_1\mathcal{B}_T)}, \quad \|f\|_{\mathcal{C}_b^1} := \|f\|_{\mathcal{C}_b^1(\pi_1\mathcal{B}_T)}, \quad \|f\|_{\mathcal{C}_b^2} := \|f\|_{\mathcal{C}_b^2(\pi_1\mathcal{B}_T)}.$$

Existence and uniqueness

Then, we can also apply Lemma 1 and Gubinelli's theorem to (Ξ, Ξ') and obtain the following estimates:

$$\|\Xi, \Xi'\| \leq CM \|f\|_{C_b^2} (|Y'_0| + \|Y, Y'\|_{X, 2\alpha}). \quad (2)$$

$$\begin{aligned} \left\| \int_0^\cdot \Xi_s d\mathbf{X}_s, \Xi \right\|_{X, 2\alpha} &\leq \|\Xi\|_\alpha + \|\Xi'\|_\infty \|\mathbb{X}\|_{2\alpha} \\ &\quad + C (\|\mathbf{X}\|_\alpha \|R^\Xi\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|\Xi'\|_\alpha) \\ &\leq \|\Xi\|_\alpha + C (|\Xi'_0| + \|\Xi, \Xi'\|_{X, 2\alpha}) (\|\mathbf{X}\|_\alpha + \|\mathbb{X}\|_{2\alpha}) \\ &\leq \|\Xi\|_\alpha + C (|\Xi'_0| + \|\Xi, \Xi'\|_{X, 2\alpha}) T^{\beta-\alpha}, \end{aligned} \quad (3)$$

where the constant C may change from line to line, depending on $\alpha, \beta, \mathbf{X}, \mathbb{X}, f(\xi)$, but uniform in T .

Existence and uniqueness

(Step 2) (Invariant ball) For $(Y, Y') \in \mathcal{B}_T$, noting that

$\|\Xi\|_\alpha = \|f(Y)\|_\alpha = \|f\|_{C_b^1} \|Y\|_\alpha$ and $|\Xi'_0| = |Df(Y_0)Y'_0| \leq \|f\|_{C_b^1}^2$, we obtain the bound

$$\begin{aligned}
 \|\mathcal{M}_T(Y, Y')\| &= \left\| \int_0^\cdot \Xi_s d\mathbf{X}_s, \Xi \right\|_{X, 2\alpha} \\
 &\leq \|\Xi\|_\alpha + C(|\Xi'_0| + \|\Xi, \Xi'\|_{X, 2\alpha}) T^{\beta-\alpha} \\
 &\leq \|f\|_{C_b^1} \|Y\|_\alpha + C \left(\|f\|_{C_b^1}^2 + CM \|f\|_{C_b^2} (|Y'_0| + \|Y, Y'\|_{X, 2\alpha}) \right) T^{\beta-\alpha} \\
 &\leq \|f\|_{C_b^1} (\|f\|_\infty + 1) T^{\beta-\alpha} + CM \left(\|f\|_{C_b^1}^2 + \|f\|_{C_b^2} (\|f\|_\infty + 1) \right) T^{\beta-\alpha},
 \end{aligned}$$

where we used $T^\alpha \leq T^{\beta-\alpha}$, $\|Y, Y'\| \leq 1$ and

$$\begin{aligned}
 |Y_{s,t}| &\leq \|Y'\|_\infty |X_{s,t}| + \|R^Y\|_{2\alpha} |t-s|^{2\alpha} \\
 &\leq (|Y'_0| + \|Y'\|_\alpha) \|X\|_\beta |t-s|^\beta + \|R^Y\|_{2\alpha} |t-s|^{2\alpha} \\
 &\leq [(\|f(\xi)\| + 1) \|X\|_\beta + 1] T^{\beta-\alpha} |t-s|^\alpha
 \end{aligned} \tag{4}$$

in the last step.

Existence and uniqueness

(Step 3) (Contraction) Setting $\Delta_s = f(Y_s) - f(\tilde{Y}_s)$ for $Y, \tilde{Y} \in \mathcal{B}_T$ and using (3), we have

$$\begin{aligned} & \|\mathcal{M}_T(Y, Y') - \mathcal{M}_T(\tilde{Y}, \tilde{Y}')\|_{X, 2\alpha} \\ &= \left\| \int_0^\cdot \Delta_s d\mathbf{X}_s, \Delta \right\|_{X, 2\alpha} \\ &\leq \|\Delta\|_\alpha + C(|\Delta'_0|_\alpha + \|\Delta, \Delta'\|_{X, 2\alpha}) T^{\beta-\alpha} \\ &\leq C\|f\|_{C_b^2} \|Y - \tilde{Y}\|_\alpha + C\|\Delta, \Delta'\|_{X, 2\alpha} T^{\beta-\alpha}. \end{aligned}$$

Then, the contraction property is obvious by taking T sufficiently small, provided that we can establish the following two estimates:

$$\begin{aligned} \|Y - \tilde{Y}\|_\alpha &\leq CT^{\beta-\alpha} \|Y - \tilde{Y}, Y' - \tilde{Y}'\|_{X, 2\alpha}, \\ \|\Delta, \Delta'\|_{X, 2\alpha} &\leq C \|Y - \tilde{Y}, Y' - \tilde{Y}'\|_{X, 2\alpha}. \end{aligned}$$

Existence and uniqueness

To obtain the first one, we replace Y by $Y - \tilde{Y}$ in (2), noting $Y'_0 = \tilde{Y}'_0$ to show that

$$\begin{aligned}\|Y - \tilde{Y}\|_\alpha &\leq \|Y - \tilde{Y}, Y' - \tilde{Y}'\|_{X, 2\alpha} \|X\|_\beta T^{\beta-\alpha} + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} T^{\beta-\alpha} \\ &\leq CT^{\beta-\alpha} \|Y - \tilde{Y}, Y' - \tilde{Y}'\|_{X, 2\alpha}.\end{aligned}$$

To obtain the second one, we write $\Delta_s = f(Y_s) - f(\tilde{Y}_s) = G_s H_s$, where

$$\begin{aligned}G_s &:= g(Y_s, \tilde{Y}_s), \quad H_s = Y_s - \tilde{Y}_s, \\ g(x, y) &:= \begin{cases} \int_0^1 Df(tx + (1-t)y)dt, & x, y \in V_T \\ 0, & x, y \notin U_T \end{cases}, \\ V_T &:= \{y \in W : \|y - \xi\| \leq \|f(\xi)\| \|X\|_\beta T^\beta + T^{2\alpha}\}, \\ U_T &:= \{y \in W : \|y - \xi\| < 2\|f(\xi)\| \|X\|_\beta T^\beta + 2T^\alpha\}.\end{aligned}$$

Existence and uniqueness

Since $f \in \mathcal{C}^3$, we may assume that g has the following regularity:

$$g \in \mathcal{C}_b^2(W \times W), \quad \|g\|_{\mathcal{C}_b^2(W \times W)} \leq C \|f\|_{\mathcal{C}_b^3(\bar{U}_T)},$$

and then employ Lemma 1 to deduce

$$\begin{aligned} (G, G') &:= (G, (D_Y g)Y' + (D_{\tilde{Y}} g)\tilde{Y}') \in \mathcal{D}_X^{2\alpha}([0, T], W \times W), \\ \|G, G'\|_{X, 2\alpha} &\leq C \|f\|_{\mathcal{C}_b^3(\bar{U}_T)} \quad \text{unif. over } (Y, Y'), (\tilde{Y}, \tilde{Y}') \in \mathcal{B}_T, \quad T \leq 1. \end{aligned}$$

Now, we claim the following inequality:

$$\begin{aligned} \|GH, (GH)'\|_{X, 2\alpha} &\leq C_{X, \xi, \alpha, f} (\|G_0\| + \|G'_0\| + \|G, G'\|_{X, 2\alpha}) \\ &\quad \times (\|H_0\| + \|H'_0\| + \|H, H'\|_{X, 2\alpha}). \end{aligned} \tag{5}$$

Note that, once we show the claim (5), we have

$$\begin{aligned}\|\Delta, \Delta'\|_{X, 2\alpha} &\leq C_{X, \xi, \alpha, f} (\|G_0\| + \|G'_0\| + \|G, G'\|_{X, 2\alpha}) \\ &\quad \times (\|H_0\| + \|H'_0\| + \|H, H'\|_{X, 2\alpha}) \\ &= C_{X, \xi, \alpha, f} (\|G_0\| + \|G'_0\| + \|G, G'\|_{X, 2\alpha}) \|H, H'\|_{X, 2\alpha} \\ &\leq C_{X, \xi, \alpha, f} \left(\|g\|_\infty + \|g\|_{C_b^1} \left(|Y'_0| + |\tilde{Y}'_0| \right) + C \|f\|_{C_b^3(\bar{U}_T)} \right) \\ &\quad \times \|Y - \tilde{Y}, Y' - \tilde{Y}'\|_{X, 2\alpha} \\ &\leq C_{X, \xi, \alpha, f} \|Y - \tilde{Y}, Y' - \tilde{Y}'\|_{X, 2\alpha},\end{aligned}$$

and conclude the contraction property.

Proof of claim: Recall that

$$X \in \mathcal{C}^\alpha([0, T], V),$$

$$Y \in \mathcal{C}^\alpha([0, T], W), \quad Y' \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, W))$$

$$G \in \mathcal{C}^\alpha([0, T], \mathcal{L}(W, \mathcal{L}(V, W))),$$

$$G' \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, \mathcal{L}(W, \mathcal{L}(V, W)))),$$

$$H \in \mathcal{C}^\alpha([0, T], W), \quad H' \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, W)).$$

Therefore, the “Leibniz rule” $(GH)' = G'H + GH'$ (or $GH' + HG'$) in our textbook is an abuse of notation, since the multiplication $G'H$ (and HG') is not well defined.

Existence and uniqueness

We substitute $G_{s,t} = G'_s X_{s,t} + R_{s,t}^G$ and $H_{s,t} = H'_s X_{s,t} + R_{s,t}^H$ to $(GH)_{s,t}$ to obtain

$$\begin{aligned}(GH)_{s,t} &= G_{s,t} H_{s,t} + G_s H_{s,t} + G_{s,t} H_s \\&= (G'_s X_{s,t} + R_{s,t}^G)(H'_s X_{s,t} + R_{s,t}^H) + G_s (H'_s X_{s,t} + R_{s,t}^H) \\&\quad + (G'_s X_{s,t} + R_{s,t}^G) H_s \\&= G_s H'_s X_{s,t} + G'_s X_{s,t} H_s + O(|t-s|^{2\alpha}).\end{aligned}\tag{6}$$

On the other hand, since $G' \in \mathcal{C}^\alpha([0, T], \mathcal{L}(V, \mathcal{L}(W, \mathcal{L}(V, W))))$ and

$$\begin{aligned}\mathcal{L}(V, \mathcal{L}(W, \mathcal{L}(V, W))) &\simeq \mathcal{L}(V \otimes W, \mathcal{L}(V, W)) \\&\simeq \mathcal{L}(W, \mathcal{L}(V, \mathcal{L}(V, W))),\end{aligned}$$

we denote the corresponding natural isomorphism

$$\iota : \mathcal{L}(V, \mathcal{L}(W, \mathcal{L}(V, W))) \rightarrow \mathcal{L}(W, \mathcal{L}(V, \mathcal{L}(V, W))),$$

so that

$$G'XH = \iota G'HX.$$

Now, the Gubinelli derivative of GH is now given by $\iota G'H + GH'$, and therefore

$$\begin{aligned}\|(GH)'\|_{\alpha} &\leq \|\iota G'H\|_{\alpha} + \|GH'\|_{\alpha} \\ &\leq \|\iota G'\|_{\infty} \|H\|_{\alpha} + \|\iota G'\|_{\alpha} \|H\|_{\infty} + \|G\|_{\alpha} \|H'\|_{\infty} + \|G\|_{\infty} \|H\|_{\alpha} \\ &= \|G'\|_{\infty} \|H\|_{\alpha} + \|G'\|_{\alpha} \|H\|_{\infty} + \|G\|_{\alpha} \|H'\|_{\infty} + \|G\|_{\infty} \|H\|_{\alpha}.\end{aligned}\tag{7}$$

Existence and uniqueness

We then substitute

$$\|G'\|_\infty \leq \|G'_0\| + \|G'\|_\alpha T^\alpha \leq \|G'_0\| + \|G'\|_\alpha,$$

$$\|G\|_\alpha \leq \|G'\|_\infty \|X\|_\alpha + \|R^G\|_{2\alpha} T^\alpha \leq (\|G'_0\| + \|G'\|_\alpha) \|X\|_\alpha + \|R^G\|_{2\alpha},$$

$$\|G\|_\infty \leq \|G_0\| + \|G\|_\alpha T^\alpha \leq \|G_0\| + (\|G'_0\| + \|G'\|_\alpha) \|X\|_\alpha + \|R^G\|_{2\alpha},$$

and similar estimate for H to (7) to conclude

$$\begin{aligned} \|(GH)'\|_\alpha &\leq C_{X,\xi,\alpha,f} (\|G_0\| + \|G'_0\| + \|G, G'\|_{X,2\alpha}) \\ &\quad \times (\|H_0\| + \|H'_0\| + \|H, H'\|_{X,2\alpha}). \end{aligned} \tag{8}$$

Existence and uniqueness

On the other hand, from the equation (6), we have

$$R_{s,t}^{GH} = (G'_s X_{s,t} + R_{s,t}^G)(H'_s X_{s,t} + R_{s,t}^H) + G_s R_{s,t}^H + R_{s,t}^G H_s, \quad (9)$$

and therefore

$$\begin{aligned} \|R^{GH}\|_{2\alpha} &\leq \|G'\|_{\infty} \|H'\|_{\infty} \|X\|_{\alpha}^2 \\ &\quad + (\|R^G\|_{2\alpha} \|H'\|_{\infty} + \|R^H\|_{2\alpha} \|G'\|_{\infty}) \|X\|_{\alpha} T^{\alpha} \\ &\quad + (\|G\|_{\infty} \|R^H\|_{2\alpha} + \|H\|_{\infty} \|R^G\|_{2\alpha}) \\ &\leq C_{X,\xi,\alpha,f} (\|G_0\| + \|G'_0\| + \|G, G'\|_{X,2\alpha}) \\ &\quad \times (\|H_0\| + \|H'_0\| + \|H, H'\|_{X,2\alpha}). \end{aligned} \quad (10)$$

Finally, we combine (8) and (10) to deduce the claim (5), and therefore the contraction property.

Existence and uniqueness

- From the Contraction mapping theorem, there exists a unique fixed point (Y, Y') of $\mathcal{M}_T : \mathcal{D}_X^{2\alpha} \rightarrow \mathcal{D}_X^{2\alpha}$ for sufficiently small T , and it is indeed contained in $\mathcal{D}_X^{2\beta}$.
- In particular, if $f \in \mathcal{C}_b^3$, the constant $C_{X,\xi,\alpha,f}$ is uniformly bounded regardless of initial point.
- Therefore, we can iterate the above procedure infinitely many times to obtain the global-in-time solution.

Regularity of the Itô-Lyons map

Notations

We first recall several ‘distance’ functionals introduced in previous chapters.

- For a rough path $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$, we define the α -Hölder rough path norm

$$|||\mathbf{X}|||_\alpha := \|X\|_\alpha + \sqrt{\|\mathbb{X}\|_{2\alpha}}.$$

- For given rough paths $\mathbf{X}, \mathbf{Y} \in \mathcal{C}^\alpha([0, T], V)$, we define the α -Hölder rough path metric

$$\varrho_\alpha(\mathbf{X}, \mathbf{Y}) := \sup_{s \neq t \in [0, T]} \frac{|X_{s,t} - Y_{s,t}|}{|t - s|^\alpha} + \sup_{s \neq t \in [0, T]} \frac{|\mathbb{X}_{s,t} - \mathbb{Y}_{s,t}|}{|t - s|^{2\alpha}}.$$

- For rough path $\mathbf{X}, \tilde{\mathbf{X}} \in \mathcal{C}^\alpha$ with $(Y, Y') \in \mathcal{D}_{\tilde{X}}^{2\alpha}$, $(\tilde{Y}, \tilde{Y}') \in \mathcal{D}_{\tilde{X}}^{2\alpha}$, we define

$$d_{X, \tilde{X}, 2\alpha}(Y, Y'; \tilde{Y}, \tilde{Y}') := \|Y' - \tilde{Y}'\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha}.$$

Stability of rough integration

We now introduce the stability of rough integration and composition discussed in chapter 4 and 7.

Lemma (Stability of rough integration, Thm 4.16)

For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, consider $\mathbf{X} = (X, \mathbb{X})$, $\tilde{\mathbf{X}} = (\tilde{X}, \tilde{\mathbb{X}}) \in \mathcal{C}^\alpha$, $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, $(\tilde{Y}, \tilde{Y}') \in \mathcal{D}_{\tilde{X}}^{2\alpha}$ in a bounded set, in the sense

$$|Y'_0| + \|Y, Y'\|_{X, 2\alpha} \leq M, \quad \varrho_\alpha(0, \mathbf{X}) \equiv \|X\|_\alpha + \|\mathbb{X}\|_{2\alpha} \leq M,$$

with identical bounds for $\tilde{X}, \tilde{Y}$ for some $M < \infty$. Define

$$(Z, Z') := \left(\int_0^\cdot Y d\mathbf{X}, Y \right) \in \mathcal{D}_X^{2\alpha},$$

and similarly for $(\tilde{Z}, \tilde{Z}')$. Then, we have

$$d_{X, \tilde{X}, 2\alpha}(Z, Z', \tilde{Z}, \tilde{Z}') \leq C \left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |Y'_0 - \tilde{Y}'_0| + T^\alpha d_{X, \tilde{X}, 2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') \right),$$

$$\|Z - \tilde{Z}\|_\alpha \leq C \left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |Y_0 - \tilde{Y}_0| + |Y'_0 - \tilde{Y}'_0| + T^\alpha d_{X, \tilde{X}, 2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') \right),$$

where $C = C(M, \alpha)$ is a suitable constant.

Stability of composition

Lemma (Stability of composition, Thm 7.5)

For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, consider $\mathbf{X} = (X, \mathbb{X})$, $\tilde{\mathbf{X}} = (\tilde{X}, \tilde{\mathbb{X}}) \in \mathcal{C}^\alpha$, $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, $(\tilde{Y}, \tilde{Y}') \in \mathcal{D}_{\tilde{X}}^{2\alpha}$ and $\varphi \in C_b^3$. Define

$$(Z, Z') := (\varphi(Y), D\varphi(Y)Y') \in \mathcal{D}_X^{2\alpha},$$

and similarly for $(\tilde{Z}, \tilde{Z}')$. Then, we have

$$d_{X, \tilde{X}, 2\alpha}(Z, Z', \tilde{Z}, \tilde{Z}') \leq C_M \left[\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |Y_0 - \tilde{Y}_0| + |Y'_0 - \tilde{Y}'_0| + d_{X, \tilde{X}, 2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') \right],$$

as well as

$$\|Z - \tilde{Z}\|_\alpha \leq C_M \left(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |Y_0 - \tilde{Y}_0| + |Y'_0 - \tilde{Y}'_0| + d_{X, \tilde{X}, 2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') \right),$$

where $C_M = C(M, T, \alpha, \varphi)$ is a suitable constant.

Stability of the Itô-Lyons map

Theorem (Rough path stability of the Itô-Lyons map)

Let $f \in \mathcal{C}_b^3$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $(Y, f(Y)) \in \mathcal{D}_X^{2\alpha}$ be the unique RDE solution to

$$dY = f(Y)d\mathbf{X}, \quad Y_0 = \xi \in W.$$

Similarly, let $(\tilde{Y}, f(\tilde{Y}))$ be the RDE solution driven by $\tilde{\mathbf{X}}$ and started at $\tilde{\xi}$ where $\mathbf{X}, \tilde{\mathbf{X}} \in \mathcal{C}^\alpha$. Then, assuming

$$\|\mathbf{X}\|_\alpha, \|\tilde{\mathbf{X}}\|_\alpha \leq M < \infty,$$

we have the local Lipschitz estimates

$$d_{X, \tilde{X}, 2\alpha}(Y, f(Y); \tilde{Y}, f(\tilde{Y})) \leq C_M \left(|\xi - \tilde{\xi}| + \varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) \right),$$

and also

$$\|Y - \tilde{Y}\|_\alpha \leq C_M \left(|\xi - \tilde{\xi}| + \varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) \right),$$

where $C_M = C(M, \alpha, f)$ is a suitable constant.

Stability of the Itô-Lyons map

Proof. Recall that the RDE solution $(Y, f(Y))$ is constructed as the unique fixed point of

$$\mathcal{M}_T(Y, Y') := (Z, Z') := \left(\xi + \int_0^\cdot f(Y_s) d\mathbf{X}_s, f(Y) \right)$$

and similarly for $(\tilde{Y}, \tilde{Y}')$. Then, by the stability of the rough integral for the integrand $(\Xi, \Xi') := (f(Y), f(Y)')$, we obtain

$$\begin{aligned} d_{X, \tilde{X}, 2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') &= d_{X, \tilde{X}, 2\alpha}(Z, Z', \tilde{Z}, \tilde{Z}') \\ &\lesssim \varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}| + T^\alpha d_{X, \tilde{X}, 2\alpha}(\Xi, \Xi', \tilde{\Xi}, \tilde{\Xi}'). \end{aligned}$$

On the other hand, by the stability of the composition, uniform in $T \leq 1$,

$$d_{X, \tilde{X}, 2\alpha}(\Xi, \Xi', \tilde{\Xi}, \tilde{\Xi}') \lesssim \varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}| + d_{X, \tilde{X}, 2\alpha}(Y, f(Y), \tilde{Y}, f(\tilde{Y}'))$$

Stability of the Itô-Lyons map

Then, we combine the above two estimates and obtain

$$d_{X,\tilde{X},2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') \leq C \left[\varrho_{\alpha}(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}| + T^{\alpha} d_{X,\tilde{X},2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') \right],$$

where $C = C(\alpha, M, f)$ is a suitable constant. Then, by taking T smaller if necessary, we may assume

$$CT^{\alpha} \leq \frac{1}{2},$$

from which it follows that

$$d_{X,\tilde{X},2\alpha}(Y, Y', \tilde{Y}, \tilde{Y}') \leq 2C \left(\varrho_{\alpha}(\mathbf{X}, \tilde{\mathbf{X}}) + |\xi - \tilde{\xi}| \right),$$

which is precisely the required bound, and these bounds can be iterated to cover a time interval of arbitrary fixed length.

Summary

- The RDE

$$Y_t = \xi + \int_0^t f(Y_s) d\mathbf{X}_s$$

admits a unique solution locally for sufficiently regular f . In particular, if $f \in \mathcal{C}_b^3$, the solution Y exists global in time.

- The Itô-Lyons map has a certain regularity (e.g. stability with respect to starting point).

THANK YOU FOR LISTENING!